

Muckund Sharma

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Mechanical Engineering

cGPA: 3.98/4.0 • President's Scholarship of Distinction (\$5,000), Mechanical Engineering Leadership Award (\$1,500)

Waterloo, ON

Sep. 2025 – Apr. 2030

TECHNICAL SKILLS

CAD, Simulation & Analysis: SolidWorks (CSWA Certified), AutoCAD, CATIA 3DEXPERIENCE (Mechanical Designer Certified), ANSYS Mechanical/Fluent, XFLR5, XFOIL, FEA, CFD, GD&T, DFM/DFA, Tolerance Analysis

Manufacturing & Fabrication: CNC Machining (Milling, Lathe), CNC Waterjet, Manual Mill & Lathe, Laser Cutting, 3D Printing (FDM), Composite Layup (Vacuum Infusion), Soldering

Programming & Electronics: Python, C++, MATLAB (Onramp Certified), Arduino, RC Systems, Brushless Motors/ESCs

Languages: English (Fluent), French (Intermediate, DELF B1 Certified)

EXPERIENCE

Mechanical Engineer

Sep. 2026 – Present

Waterloo Aerial Robotics Group (WARG)

Waterloo, ON

- Engineered the Pegasus 1 landing gear in SolidWorks (127.8 g per leg): carbon-fiber tubes and a 3D-printed PETG-CF bracket with an intentional breakaway neck that fails before the airframe in a hard crash.
- Validated the bracket with orthotropic linear-static FEA in ANSYS across a 1–4g sweep: 3.81 FOS at 1g, tensile failure onset at the breakaway neck at 4g, and mesh convergence confirmed (1.22% peak-stress change).
- Conducted a wingtip trade study for the team's 2.42 m UAV wing, modeling induced against added profile drag for five devices: none changed wing drag by more than 5% (0.17 N), consistent with NASA Langley wind-tunnel data.
- Designed the recommended raked tip in SolidWorks (131 mm long, 200 to 90 mm chord, 54° sweep, 0–10° dihedral; accepted by the team) and defined its XFLR5 VLM validation plan.
- Designed a 3D-printable protective enclosure for an APD 120F3 ESC with M3 mounts, component clearances, passive ventilation, and terminal protection.

Mechanical Engineer

Sep. 2025 – Aug. 2026

Waterloo Rocketry

Waterloo, ON

- Contributed airframe materials testing to Polaris, which set the amateur liquid-fuel altitude world record (63,497 ft, Launch Canada 2026): compression-tested six fiberglass laminates to ASTM D695 / DIN EN 2850, and the tested Airstone epoxy laminate was adopted for the airframe.
- Built a Python pipeline turning load/displacement data into stress-strain curves to compare all six laminates.
- Machined a Boeing BSS 7260 test fixture (manual mill, lathe) and CNC-waterjet aluminum alignment tabs; executed a full vacuum-infusion composite layup end to end.

Project Coordinator & Teaching Assistant

May 2026 – Aug. 2026

York Region Educational Services (YRES)

Markham, ON

- Mentored and evaluated 6 summer staff, and launched a Jira workspace for a 200+ participant volunteer event, giving the team real-time progress visibility for the first time.

PROJECTS

MX-01 RC Trainer: Design, Aerodynamic Analysis and Flight Test

Aug. 2026 – Present

- Designed a Cessna 172-inspired RC trainer solo in SolidWorks and hand-built it from EPS foam (0.79 m span, 0.6 kg) for about \$140; selected, wired, and commissioned the propulsion and control system.
- Chose a Clark Y airfoil from XFOIL polars (Re 50k–150k) and modeled the wing in XFLR5: peak L/D about 15 at 4°, L/D about 14 at the 10 m/s cruise point; set a 2.3° mounting incidence.
- Flight-tested over 12 flights (about 1 hr airtime), reducing motor vibration with a second motor mount and softening over-sensitive controls with 30% transmitter expo.

VEX IQ Automated Coin Management System

Jan. 2026 – Apr. 2026

- Built a multi-user coin ATM (laser-cut frame, 3D-printed parts), managing build and software for a 4-person team, with a 700+ line C++ controller (7 modes, about 50 Hz).
- Achieved 96.7% coin-detection accuracy, 100% withdrawal accuracy, and a 0.41 s average jam-detection response.

Mechanical Keychain Assembly

Nov. 2025 – Dec. 2025

- Modeled a four-part aluminum assembly in SolidWorks with GD&T and datum-controlled drawings, then machined it on manual and CNC equipment and confirmed fit-up across all interfaces.

Concrete T-Beam Design and Fabrication

Jan. 2025 – May 2025

- Designed an inverted-T reinforced concrete beam using over 20% less concrete than a rectangular section, with variable-spaced stirrups per ACI 318-19, an AutoCAD drawing, and custom formwork.